CHAPTER 7

INDEX NUMBERS

CHAPTER ESSENCE AT A GLANCE:
Measure economic barometers: price relatives, simple aggregative index, weighted Laspeyres, Paasche, and Fisher ideal index, Consumer Price Index (CPI), and Wholesale Price Index (WPI).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Index Numbers as Economic Barometers:** Statistical devices designed to measure net changes in the general level of prices, industrial output, or cost of living relative to a base period.

• **Weighted Index Number Models:** Laspeyres uses base year quantities as weights ($P {01}^{L}$); Paasche uses current year quantities as weights ($P {01}^{P}$); Fisher's Ideal Index is the geometric mean of Laspeyres and Paasche.

• **Fisher's Ideal Index Perfection:** Fisher index is called 'Ideal' because it uses geometric mean, incorporates both base and current quantities, and satisfies both Time Reversal and Factor Reversal Tests.

• **Consumer Price Index (CPI) and Deflating:** CPI measures cost of living changes for specific socio-economic classes, used to calculate Dearness Allowance (DA) and deflate nominal wages into real wages.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$P_{01}^{L} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 \quad [\text{Laspeyres Base Weighted Index}]$$

Laspeyres Price Index Formula

$$P_{01}^{P} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 \quad [\text{Paasche Current Weighted Index}]$$

Paasche Price Index Formula

$$P_{01}^{F} = \sqrt{P_{01}^{L} \times P_{01}^{P}} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

Fisher Ideal Geometric Price Index

$$\text{Real Wage} = \frac{\text{Nominal Money Wage}}{\text{Consumer Price Index (CPI)}} \times 100$$

Purchasing Power Deflation Equation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Laspeyres index tends to overestimate price rises because it uses base quantities, ignoring consumer substitution toward cheaper goods.

• **Alert:** Paasche index tends to underestimate price rises because it uses current quantities heavily weighted toward goods that have become relatively cheaper.

• **Alert:** Fisher's Ideal Index satisfies Time Reversal Test (P01 * P10 = 1) and Factor Reversal Test (P01 * Q01 = V01).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Index Numbers. Why are they called 'Economic Barometers'? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Specialized statistical averages designed to measure relative changes in price levels, production, or economic variables over time. **[1 Mark]**

• **Economic Barometer:** Just as a barometer measures atmospheric pressure, index numbers measure fluctuations in the economic atmosphere like inflation and purchasing power. **[1 Mark]**

2. Why is Fisher's Index Number called the 'Ideal' Index Number? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Reason 1:** Based on geometric mean, which is the mathematically best average for constructing index numbers. **[1 Mark]**

• **Reason 2:** Balances base year and current year weights, satisfying both Time Reversal and Factor Reversal Tests. **[1 Mark]**

3. Distinguish between Consumer Price Index (CPI) and Wholesale Price Index (WPI). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Stage of Exchange:** WPI measures price changes at the wholesale transaction level; CPI measures retail prices paid by ultimate end consumers. **[1.5 Marks]**

• **Inclusion of Services:** WPI tracks physical goods only, excluding services; CPI includes retail services like housing, education, and transport. **[1.5 Marks]**

4. A worker earned Rs 20,000 per month in the base year 2015. In 2024, his monthly salary increased to Rs 36,000, while the Consumer Price Index rose to 200. Calculate his Real Wage in 2024 and determine whether his purchasing power increased or decreased. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Real Wage Formula:** Real Wage = (Nominal Money Wage divided by CPI) * 100. **[1 Mark]**

• **2024 Real Wage:** (Rs 36,000 divided by 200) * 100 = Rs 18,000. **[1 Mark]**

• **Purchasing Power Verdict:** Real wage declined from Rs 20,000 to Rs 18,000; his purchasing power decreased by Rs 2,000 despite money wage rising. **[1 Mark]**

5. State any three difficulties encountered in the construction of an Index Number of prices. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Choice of Base Year:** Selecting a normal base year free from abnormal events like wars, famines, or hyperinflation is difficult. **[1 Mark]**

• **Choice of Representative Commodities:** Difficult to choose a standard basket of goods that accurately reflects consumption patterns across different income classes. **[1 Mark]**

• **Changing Quality Over Time:** Products undergo rapid technological improvements, making price comparisons over decades misleading. **[1 Mark]**

6. Given Sum of p1*q0 = 500, Sum of p0*q0 = 400, Sum of p1*q1 = 660, Sum of p0*q1 = 550. Calculate: (a) Laspeyres Price Index, (b) Paasche Price Index, (c) Fisher Ideal Price Index. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Laspeyres Index:** (500 divided by 400) * 100 = 125.0. **[1 Mark]**

• **(b) Paasche Index:** (660 divided by 550) * 100 = 120.0. **[1 Mark]**

• **(c) Fisher Ideal Index:** sqrt(125.0 * 120.0) = sqrt(15,000) = 122.47. **[1 Mark]**

7. Explain the Time Reversal Test and the Factor Reversal Test of Index Numbers. Prove that Fisher's Index Number satisfies both tests. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Time Reversal Test:** Reversing the time subscripts (0 and 1) should yield numbers that are reciprocal of each other: P01 * P10 = 1. Fisher index algebraically cancels out all terms, satisfying it completely. **[2 Marks]**

• **Factor Reversal Test:** Interchanging price (p) and quantity (q) should yield a value ratio: P01 * Q01 = (Sum of p1*q1) divided by (Sum of p0*q0) = V01. Fisher index algebraically equals the exact value ratio, satisfying it completely. **[2 Marks]**